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Ysbyty Gwynedd Anaesthetic Department

Acute pain management in opioid tolerant patients

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1. INTRODUCTION

When individuals have been taking opioids for a prolonged period (illegally or prescribed), effective titration of analgesia can be difficult. These patients will be tolerant to and physically dependent on these drugs, and some may have an addiction to opioids.

The prescription of opioids, particularly for the treatment of chronic noncancerous pain, has risen sharply in many developed countries of the world (Huxable *et al*, 2011). There has also been an associated increase in the use of opioids for nonmedical purposes.

The management of effective analgesia in these patients can be a complex task and may require treatment over longer periods and significant deviation from standardised protocols.

Definitions

Tolerance	A decrease in sensitivity to opiates resulting in less effect from the same dose, or the need for progressively larger doses to achieve the same effect
Opioid-induced hyperalgesia	The development of an increased sensitivity to pain associated with long term opioid use
Physical dependence	A physiological adaptation to a drug characterised by withdrawal, if a drug is suddenly stopped or reduced in dose

2. POLICY STATEMENT

BCUHB is committed to ensuring acute pain is managed appropriately in opioid tolerate inpatients by ensuring the opioids are prescribed administered and monitored appropriately.

3. AIMS

This document applies to all opioid dependent adults at Ysbyty Gwynedd, and intended for all physicians, nurses and pharmacists involved in their acute management of pain.

4. OBJECTIVES

- An adequate review and assessment prior to initiating acute pain therapies.
- Preventing opiate-withdrawal.
- Involvement of multidisciplinary and specialist teams (substance misuse, psychiatrists and general practitioners) and treating co-morbidities (e.g. depression and other mental health illnesses).
- Organising appropriate management on discharge
  (ANZCA and FPM, 2015, Huxable *et al*, 2011).

5. SCOPE

Ysbyty Gwynedd clinical staff involved in the care of opioid tolerant patients who undergo additional training provided by the Acute Pain Service.

6. ROLES AND RESPONSIBILITIES

Anaesthetist
• Overall responsibility for perioperative analgesia
Surgeon
• Ongoing postoperative monitoring and care on the ward
Acute Pain Team
• Daily review of post operative pain management • Advice and support • Liaise with Key Workers and Substance Misuse Teams
Nursing Ward Staff
• Provide adequate post-operative monitoring of patient
Pharmacy
• Clarify preoperative medications • Liaise with Key workers and Substance Misuse

7. MAIN BODY

Opioid tolerance and opioid-induced hyperalgesia

- Patients on long-term opioid therapy may have developed a tolerance to the drug, which means high doses are needed to provide good analgesia. Opioid-induced hyperanalgesia (OIH) may also be present. This will have an opposite effect and increased doses will not improve the patient's pain relief. After

continuous exposure to an opioid, tolerance is probably seen after 10-14 days. (Schug, 2012).

- Patients with high opioid requirement for the treatment of acute pain for a week or two are at risk of withdrawal if stopped abruptly. Dose reductions of 20-25% daily will allow opiate dose reduction without withdrawal. A more rapid reduction and management of withdrawal can be given if the patient is given clonidine (ANZCA & FPM, 2015).
- If opioid analgesia is inadequate in patients receiving additional opioids for the management of their acute pain, patient tolerance should be assumed as long as there are no other identifiable reasons (e.g., postoperative complications). In this situation, opioid doses should be increased appropriately and and/or multi-analgesia considered.

- Withdrawal in opioid tolerant-patients can be prevented if their usual opioid (or equivalent) is continued.

- Withdrawal from opioids is characterised by abdominal cramping, aches and pains, insomnia, dysphoria, anxiety, nausea and vomiting, yawning, sweating, lacrimation, rhinorrhoea, anxiety, restlessness, dilated pupils, piloerection (goose flesh), chills, tachycardia and hypotension hypertension, (ANZCA and FPM, 2015).

- In patients with physiological opioid dependence, withdrawal may occur within 4-6 hours after the last dose of short-aching opioid, but later with methadone or slow-release preparations.

- The postoperative baseline opioid requirement is calculated from the patient's opioid requirements preoperatively. This provides an approximation of the opioid requirements only. However, the actual requirements may be more or less.

Preoperative

- A multidisciplinary approach, involving the anaesthetist, involving the chronic pain and acute pain team should be used to devise a preoperative analgesic plan. Pre operative analgesia should be assessed for efficacy and unrelieved pain treated if possible

- It is important to seek information about all usual medications (opioid and non opioid) and there doses. Questions should also be asked about over the counter medications, illicit drugs, and alcohol (Huxtable et al, 2011).

- The doses of prescribed medications should be verified by the patient's treating doctor and pharmacist. It is also worth checking if they are taken as prescribed.

- In patients with pre-existing pain, it is useful to have their pain scores and functional status. It should be explained the primary focus, is to effectively manage their acute pain episode, and not to improve the management of their chronic pain.

- Opioid-tolerant patients may have significant emotional and psychiatric co-morbidities. Therefore, management of behavioural, psychological, medical, social, and other factors may be needed in addition to analgesia. This may require the assistance of other specialist team, such as chronic pain, palliative care, substance misuse and psychiatry.

Assessment

- Consider the psychological, social and pathological, triggers for pain deterioration in chronic pain patients (Quinlan, 2012)
- There are specific factors to be sought in all opioid-tolerant patients, chronic and cancer pain and those with substance misuse:

Opioid-tolerant patients	**Additional information Continuous peripheral nerve blockade (CPNB) or cancer patients**	**Additional information in substance misuse patients**
Current providers of treatment	Pain diagnosis	Opioid substitution and doses (methadone and buprenorphine)
Opioid and nonopioid medications	Usual pain scores	Other prescribed or illicit substances
Dose verification of all other medications	Functional status	Route of administration
Non-prescribed drugs (illicit drugs, nicotine and alcohol)	Prognosis (cancer pain)	Registered prescriber and dispensing pharmacy
Allergies and reactions	Psycho spiritual issues (e.g. anxiety, depression, coping strategies and end-of –life issues	Medical and psychiatric co morbidities
Acute pain management expectations		

Analgesia

- A multimodal analgesic approach will be of most benefit and the use of non-opioid (paracetamol and NSAIDs, drugs that may attenuate hyperalgesia and tolerance (ketamine and gabapentinoids), and regional analgesic techniques should be maximised in this type of patient. However, opiates will be the mainstay of the acute pain management of moderate to severe pain. Tramadol may also be used, although is sole use instead of opiates is not recommended as it may prevent opioid withdrawal.

- Opioid requirement will be often higher than average. It is difficult to judge but can be three times greater than opioid naive patients (Rapp et al, 1995). The dose should account for the patients current opioid requirements, although difficult as often illicit drugs are involved. Start with a conservative estimate and rapidly titrate until satisfactory analgesia is achieved, or until the dose is limited by side effects. If high dose requirements are expected, it is easier and more effective to deliver via PCA. A background infusion can be used to deliver the equivalent usual opioid dose if there is no oral route.

- Pain scores tend to be higher in opioid-tolerant patients, especially those with chronic noncancerous pain (ANZCA and FPM, 2010), and therefore not a reliable guide. An objective assessment of function, such as the ability to cough or mobilise after a laparotomy or fractured ribs. The comparison with preadmission pain scores is also useful. A patient with pinpoint pupils means the drug is exerting near maximal effect in the CNS and no safe to increase and other strategies will need to be utilised.

Tolerance and hyperanalgia

Strategies to attenuate opioid tolerance and hyperalgia and may improve analgesia in the acute pain setting:

a) NMDA-receptor antagonists
b) Opioid-receptor antagonists
c) Opioid rotation
d) Other adjuvant medicines

a) NMDA-receptor antagonists

- The NDMA receptor is involved in the development of tolerance and OIH (Chang, 2007a). In patients on long-term opiates, ketamine had led to improve pain relief and reduce opioid requirements (Mitra, 2008).

- Ketamine, magnesium and amantadine reduce tolerance and OIH associated with remifentanyl use.

b) Opioid-receptor antagonists

Ultra-low doses of naloxone have been shown to attenuate opioid tolerance (Wang, 2005) and remifetanil-induced OIH (Aguado 2013).

c) Opioid Rotation

- Switching from one opiate to another is common practice in palliative care. The aim is to reduce opioid related side-effects and improve pain relief. It is thought differing receptor activity and incomplete cross-tolerance is likely to exist, so when an opiate is changed, the “new” opioid may be more effective when used less than equianalgesic dose (Huxable et al, 2011). Commonly, the “new” opioid is started at 30-50% of previous dose (methadone is the exception when much lower doses are used).

- In the acute pain setting, switching to another opioid because of inadequate pain relief is best left until increased opioid doses and other analgesic strategies have been tried.

d) Other adjuvants medicines

Are primarily used for their antitolerance, antiallodynic and antihyperalgesia effects (Huxtable, 2011).

- There is evidence that ketamine can reduce opioid requirements and improve pain relief in opiate-tolerant patients (ANZA and FPM, 2015, Lofus et al, 2010). Is has also been shown to have anti hyperanalgesic properties.

- Ketamine at low doses 50-200mg/24hours or an infusion rate of 0.1 mg/kg/hr by IV or subcutaneous is useful, with few side effects. It is more useful in patients on higher opiate doses prior to admission.

- Gabapentin has been shown to attenuate opioid tolerance (Aguado, 2012) and OIH (Wei, 2012). Their ability to reduce anxiety may also benefit some patients (Baldwin *et al*, 2013).

Patient Controlled Analgesia (PCA)

- A useful way of delivering opioids in opioid-tolerant patients, especially when high requirements are needed. Additionally, if patients are unable to continue their long-term opioid medication (e.g. nil by mouth), a background infusion can be used to cover this baseline requirement.

- It is difficult to estimate opioid requirement in tolerant patients. One method is to base the bolus dose on the patient's pre-admission opioid requirement. If on methadone regularly, it is more difficult because of the variable half-life and it long duration of action. In practice a " conversion" of:

  1mg oral methadone = 1mg parenteral morphine

  Is used initially and adjusted to suit. This is a working estimate only and not based on equianalgesic doses.

- The PCA bolus dose should be titrated according to the patient's response. High PCA doses should be managed by The Acute Pain Team (Bleep 115) managed in an environment where nurses and medical staff are appropriately trained, and adequate monitoring is available.

- When a patient is tolerating oral fluids, they can restart their usual opioids medication to replace the background infusion. High PCA dose may delay a patient being managed on oral opioids alone.

Discharge

- Patients prescribed opioids after discharge for acute pain management should be advised not to drink alcohol or take sedative medications while taking these medications. Patient who have recently starting opioid medications are at more risks of falls (Soderberg *et al*, 2013).

- Patients prescribed opiates for acute pain ongoing after discharge, for as long as two years following surgery, for a variety of reasons (Macintyre *et al*, 2014).

- Long-term opiate use initially prescribed for acute pain short term may be due to psychological factors such as depression rather than pain (Macintyre et al, 2014). Persistent postsurgical pain has been linked with depression, anxiety, pain catastrophizing (Theunissen, 2013).

- It is important to recognise the risk factors, and follow up the patient following discharge. Patients requiring opioids for acute pain months after discharge need further evaluation. Pain persisting longer than three months is classed as chronic not acute, requiring alternative treatment strategies.

- The increasing problems of opioid misuse are well recognised. There has been a rapid rise in the in the prescription of opioids (Manchikanti *et al,* 2012), coincided with the increase in deaths relating to the non medical use of opiates, and hospital admissions due to complications from opiates. Around 50% of those obtaining opiates for nonmedical use are sourced from their family and friends (Stafford & Burns).

At risk patients

- Before prescribing opiates on discharge the risk of misuse must be considered. A tool that screen for potential misuse is the opioid risk tool (ORT) (Appendix 1) (Webster & Webster, 2005). A total score of 8 is high risk. It may be less sensitive than others, but quick and easy to use in the acute pain setting.

Choice of opiates

- Immediate release opiates are recommended (Thorsen *et al*, 2014). Methadone is restricted to those experienced in their use (e.g. Substance Misuse).

- Generally, the choice of opioid is of less importance than the dose prescribed. For inpatients, it should be based on the requirements in the last 24 to 48 hours prior to discharge.

- It is better to have small repeated prescriptions. In some patients, “interval dispensing’ (e.g. one or two day intervals) may be needed.

Opioid tolerant patients

- A patient taking long-term opiates prior to admission for chronic pain, the aim is to minimize alterations to the patient’s usual regime. It there is a short-term increase in their opioid dose after discharge or additional opiates a required, this should be done in consultation with their doctor.

- Alterations to methadone and buprenorphine doses as part of opioid substitution programs should also be minimised.

- Generally acute pain will reduce as the patient recovers. Therefore, opioid doses should be titrated during their recovery using a “Reverse Pain Ladder” (McQuay, 2004).

Key Messages

1. Alpha-2 agonists (clonidine and lofexidine) reduce opioid-withdrawal symptoms.

2. Remifentanil use leads to opioid-induced hyperalgesia, which is attenuated by propofol, NMDA-receptor agonists and pregablin.

3. Gabapentin and pregabalin attenuate opioid-induced hyperanlgesia/tolerance and reduce opioid-withdrawal symptoms

4. In opioid-tolerant patients, ketamine improves pain relief after surgery, and may reduce opioid requirements.

5. Opioid-tolerant patients report higher pain scores, have slower pain resolution leading to longer hospital stay and increased readmissions, but have a lower incidence of opioid-induced nausea and vomiting.

6. Opioid-tolerant patients may have significantly higher opioid requirements and patient variation in the doses needed than opioid-naïve.

Conclusions based on clinical experience and expert opinion.

- ✓ Usual preadmission opioid regimens should be maintained where possible or appropriate substitutions made.

- ✓ Liaison with all health care professionals involved in the treatment of the opioid-tolerant patient is important.

- ✓ Opioid-tolerant patients are at risk of opioid withdrawal if non-opioid analgesic regimens or tramadol or tapentadol alone are used.

- ✓ PCA settings may need to include a background infusion to replace the usual opioid dose and a higher bolus dose.

- ✓ Adjuvants are used for their antitolerance, antihyperalgesic, and antiallodynic effects and there is some evidence upon which to base the choice of agent.

- ✓ In patients with escalating opioid requirements, management considerations are the development of tolerance or opioid-induced hyperalgesia).

- ✓ Long-term opioid use may increase the risk of sleep-disordered breathing, which requires appropriate assessment, monitoring and management in the preoperative period.

- ✓ Following short-term opioid dose escalation for acute pain, a “reverse analgesic ladder” approach, using stepwise reduction to the patient’s usual opioid regimen is recommended.
  (ANZCA and FPM, 2015)

8. EQUALITY AND WELSH LANGUAGE

The equality impact assessment (EQIA) did not identify an adverse impact for any protected group. This best practice guidelines are intended as a prescribing resource; therefore, it is not anticipated to impact the use of the Welsh language.

9. WELL BEING FOR FUTURE GENERATIONS

The use of the guideline, for the indication stated in section 1, has no effect on the wellbeing of future generations.

10. ENVIROMENTAL IMPACT ASSESSMENT

An ‘Environment Impact Assessment’ has not been undertaken it is not anticipated to have an environmental impact.

11. RESOURCES

Continuing funding is required for the supply of disposable equipment and the servicing and repair of electronic devices.

12. TRAINING

Time and staffing to allow the training and education of all staff involved in the management of patients with opioid tolerance. Training records stored locally by BCUHB.

13. IMPLEMENTATION

Successful implementation will require dissemination of information to all applicable staff through the anaesthetic department intranet.

14. FURTHER INFORMATION – CLINICAL DOCUMENTATION

See Appendix 1.

15. AUDIT

Anaesthetists and Acute Pain Team will ensure guidance compliance. No audit tool available.

16. REVIEW

The policy will be reviewed and updated every three years, or sooner if required.

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Appendix 1

Opioid Risk Tool

	Tick if applicable	Female	Male
Family history of abuse			
Alcohol		1	3
Illegal drugs		2	3
Prescription drugs		4	4
Personal history of substance misuse			
Alcohol		3	3
Illegal drugs		4	4
Prescription drugs		5	5
Age			
16-45 years		1	1
History of preadolescent sexual abuse			
		3	0
Psychological disease			
Attention-deficit disorder, obsessive-compulsive disorder, bipolar, schizophrenia		2	2
Depression		1	1
Total			